Statistical power of Tree-Based Scan Statistics (TBSS) for the surveillance of pregnancy drug safety in Swedish health registers

Stamatía Tsampá[1], Julia Eriksson[1], Fredh Netterström-Wedin[1], David Hägg[1], Cathrine Axfors[1,2,3]

1. Centre for Pharmacoepidemiology, Department of Medicine Solna, Karolinska Institutet, Sweden, 2. Department of Women's and Children's Health, Uppsala University, Sweden, 3. Meta-Research Innovation Center at Stanford (METRICS), Stanford University, Stanford, CA

Background

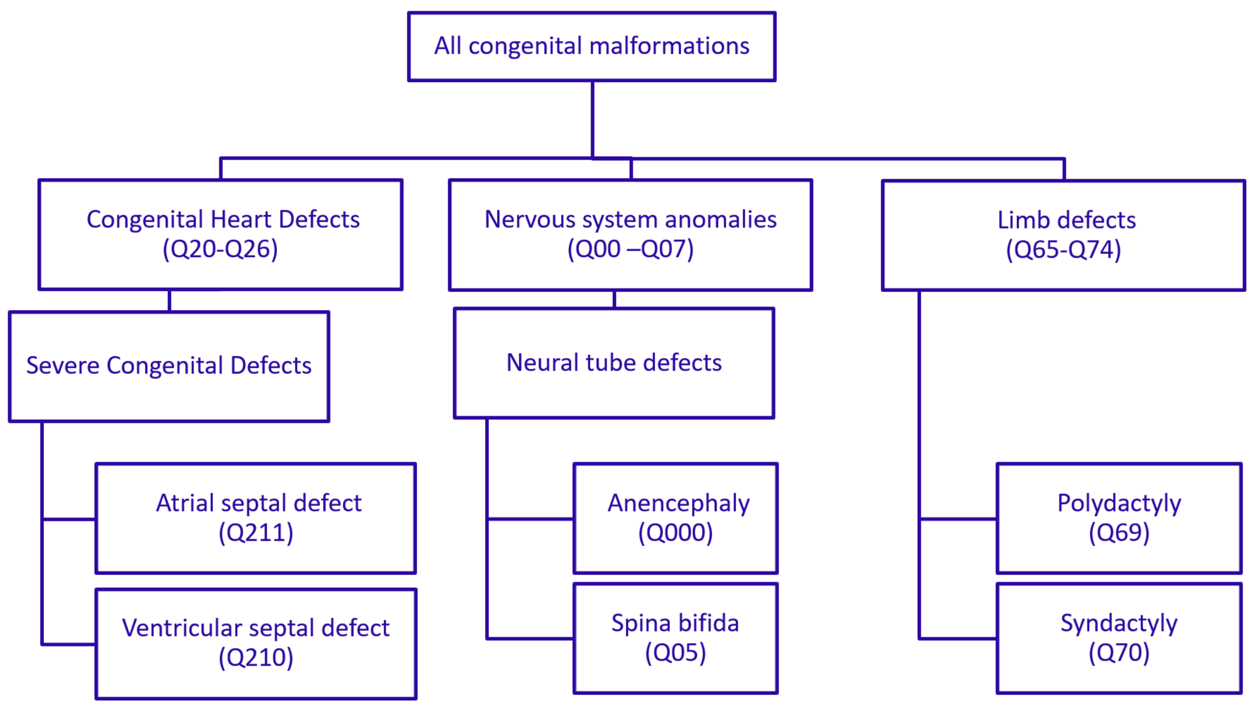
Investigations on the safety of a broad range of medications and outcomes are necessary to inform pregnant individuals on potential adverse effects. TBSS is a hypothesis-generating tool for drug safety surveillance in large-scale observational data. A TBSS model studies many related outcomes in a hierarchical tree and adjusts for multiplicity. Its statistical power, which depends on several factors, has not been studied in a Nordic context.

This study aims to investigate the statistical power of TBSS to identify alerts for pregnancy drug safety using Swedish health register data.

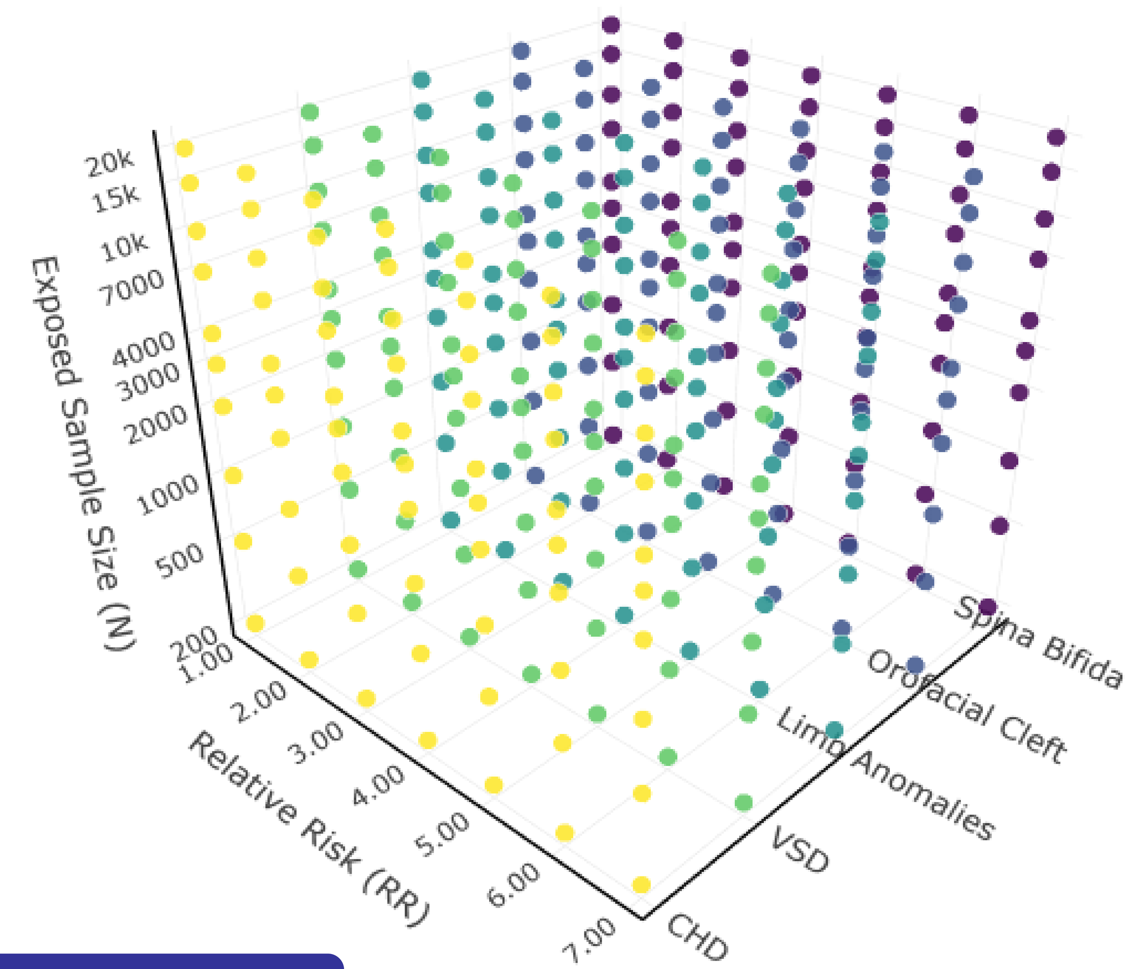
Methods

We construct a hierarchical outcome tree including all malformation outcomes as defined by EUROCAT 1.5. A nationwide cohort drawn from Swedish national register data 2006-2022 is used to calculate malformation prevalences among pregnancies that resulted in singleton births. Malformation outcomes are assessed up to one year after delivery.

Using the TBSS Poisson model, we simulated observed event counts for each outcome, for scenarios with different exposed sample sizes and outcome tree versions (full and pruned). At selected outcomes (Congenital Heart Defects (CHD), Ventricular Septal Defects (VSD), Limb Anomalies, Orofacial Clefts and Spina Bifida) we injected, separately, artificial risks with different effect sizes. Outcome-specific statistical power, for each scenario separately, is calculated as the proportion of iterations where the TBSS analysis (correctly) rejects the global null hypothesis.



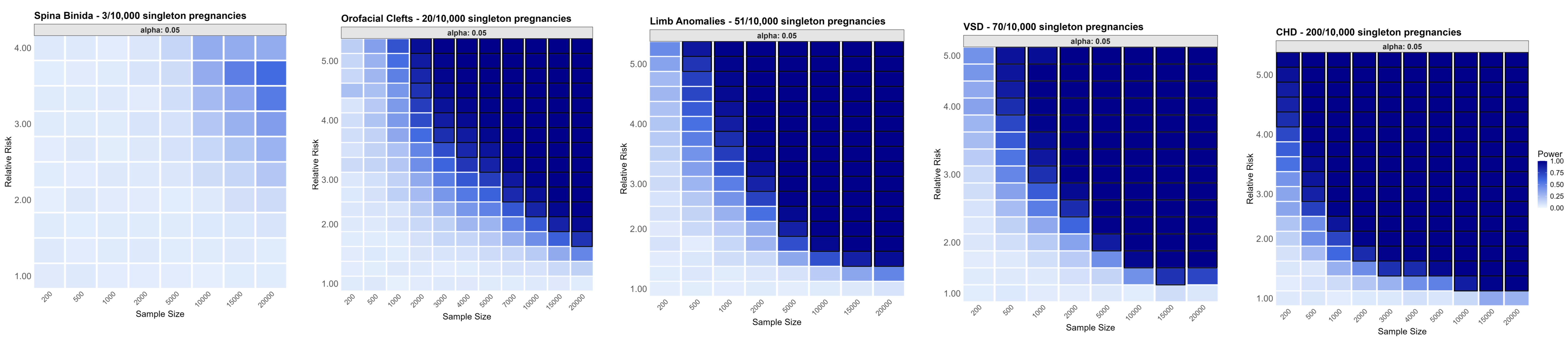
Schematic representation of the hierarchical tree used in the TBSS simulation study. For clarity, only indicative nodes are displayed rather than the full tree.

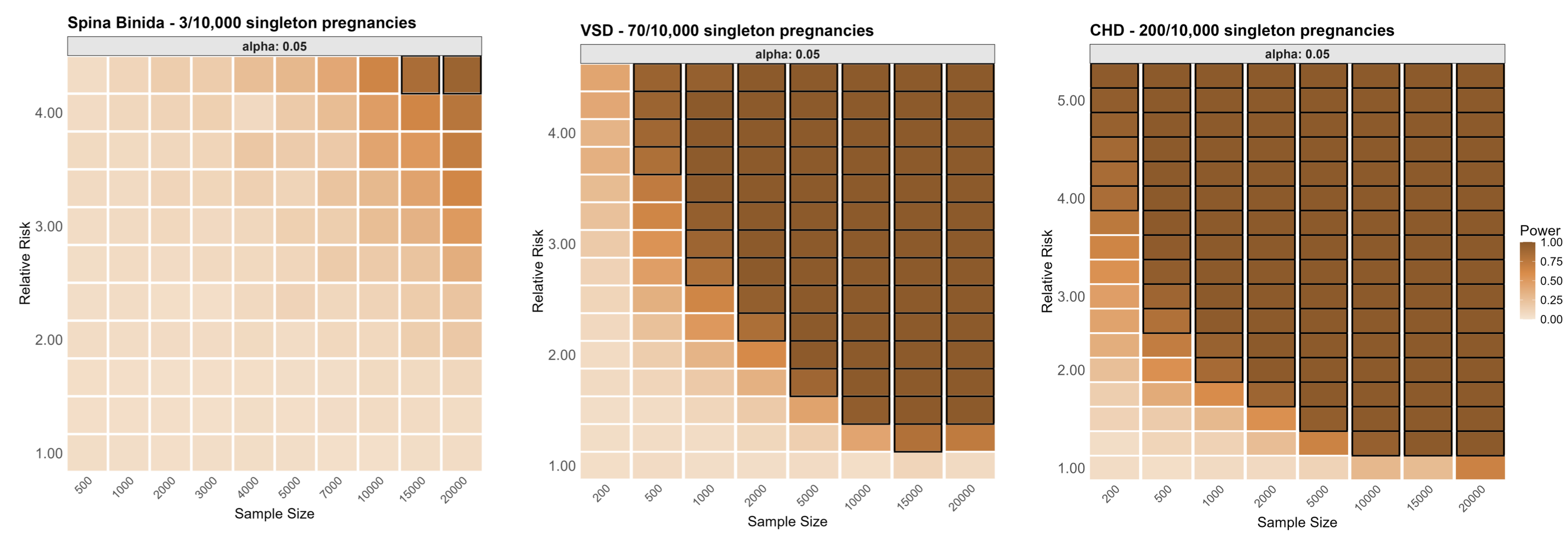


Schematic of the factorial parameter space explored by the simulation analysis. Axes: Exposed Sample Size, Relative Risk, and Outcome-Specific Nodes. The simulation was iterated across two tree versions (pruned: outcomes equaling subgroups defined by EUROCAT 1.5; full: complemented with all pertinent 4-digit ICD-10-SE codes).

Results

Statistical power was estimated across the full factorial parameter space (exposed sample size × relative risk × selected outcomes × tree version). Heatmaps display outcome-specific power for the full (blue-colored heatmaps) and pruned (orange-colored heatmaps) tree versions across selected outcomes. Bolded tiles reflect scenarios where power ≥80% was achieved.





Conclusion

Statistical power was comparable to previous analyses using an ICD-10-CM–based (United States) outcome tree. Pruning the outcome tree improves power slightly, at the cost of not investigating very granular outcomes individually. For less commonly used medicines, multinational data will often be needed to investigate uncommon subgroups.

Stamatia Tsampa, M.D., MSc
Centre for Pharmacoepidemiology,
Department of Medicine Solna, Karolinska Institutet, Sweden, stamatia.tsampa@ki.se

Karolinska Institutet